TASK 1 : Execute a Linux Command Using fork() and exec()

```
krishnasree08@krishnasree08:~$ mkdir task1
krishnasree08@krishnasree08:~$ cd task1
krishnasree08@krishnasree08:~/task1$ pwd
/home/krishnasree08/task1
krishnasree08@krishnasree08:~/task1$ nano command_exec.c
krishnasree08@krishnasree08:~/task1$ gcc command_exec.c -o command_exec
krishnasree08@krishnasree08:~/task1$ ls
command_exec  command_exec.c
krishnasree08@krishnasree08:~/task1$ ./command_exec
Enter a Linux command: ls

Parent Process:
Parent PID : 613

Child PID  : 614
Child Process:
Child PID  : 614
Parent PID : 613

Executing command...

command_exec  command_exec.c

Child process has completed.
Parent process continues execution.
```

```
krishnasree08@krishnasree08:~/task1$ ./command_exec
Enter a Linux command: pwd

Parent Process:
Parent PID : 617

Child PID  : 618
Child Process:
Child PID  : 618
Parent PID : 617

Executing command...

/home/krishnasree08/task1

Child process has completed.
Parent process continues execution.
```

The program successfully demonstrated the basic process management mechanism in Linux. The fork() system call created a child process, while execvp() was used by the child to execute the command entered by the user. The PID of both processes was displayed using getpid() and getppid(). The parent process used wait() to wait for the child process to complete. The output confirms that the command was successfully executed by the child process.

Summary: In this task, a C program was developed to understand how Linux creates and manages processes. The fork() system call was used to create a child process, and execvp() was used by the child to execute the Linux command entered by the user. The getpid() and getppid() functions were used to display the process IDs, while wait() allowed the parent process to wait for the child to finish. This demonstrated the basic process creation and command execution mechanism in Linux.

TASK 2 : Hardware Resources and Operating System Services Aim

```
krishnasree08@krishnasree08:~/task1$ uname -a
Linux krishnasree08 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
```

```
krishnasree08@krishnasree08:~/task1$ lscpu | head -20
Architecture:                            x86_64
CPU op-mode(s):                          32-bit, 64-bit
Address sizes:                           48 bits physical, 48 bits virtual
Byte Order:                              Little Endian
CPU(s):                                  16
On-line CPU(s) list:                     0-15
Vendor ID:                               AuthenticAMD
Model name:                              AMD Ryzen 7 7730U with Radeon Graphics
CPU family:                              25
Model:                                   80
Thread(s) per core:                      2
Core(s) per socket:                      8
Socket(s):                               1
Stepping:                                0
BogoMIPS:                                3992.41
Flags:                                   fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmx
ext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 s
se4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_co
re ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsave
erptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
Virtualization:                          AMD-V
Hypervisor vendor:                       Microsoft
Virtualization type:                     full
L1d cache:                               256 KiB (8 instances)
krishnasree08@krishnasree08:~/task1$ nproc
16
krishnasree08@krishnasree08:~/task1$ lsblk
NAME
    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0     2G  0 disk [SWAP]
sdd   8:48   0     1T  0 disk /mnt/wslg/distro
```

```
krishnasree08@krishnasree08:~/task1$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda  ext4   1.0
sdb  ext4   1.0
sdc  swap   1           d20d965c-2165-4314-9f35-feb4aed0dc0d                [SWAP]
sdd  ext4   1.0         19430526-c024-41cd-8fb9-a764b390a31b  953.7G     0% /mnt/wslg/distro
                                                                            /
krishnasree08@krishnasree08:~/task1$ ps
    PID TTY          TIME CMD
    309 pts/0    00:00:00 bash
    714 pts/0    00:00:00 ps
krishnasree08@krishnasree08:~/task1$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  21532 12964 ?        Ss   08:13   0:00 /sbin/init
root           2  0.0  0.0   3180  2204 hvc0     Sl+  08:13   0:00 /init
root           6  0.0  0.0   3196  2128 hvc0     Sl+  08:13   0:00 plan9 --con
root          52  0.0  0.2  66836 19544 ?        S<s  08:13   0:00 /usr/lib/sy
root         100  0.0  0.0  25284  6640 ?        Ss   08:13   0:00 /usr/lib/sy
systemd+     116  0.0  0.1  21344 13192 ?        Ss   08:13   0:00 /usr/lib/sy
systemd+     117  0.0  0.1  91036  7988 ?        Ssl  08:13   0:00 /usr/lib/sy
root         180  0.0  0.0   4244  2688 ?        Ss   08:13   0:00 /usr/sbin/c
message+     181  0.0  0.0   9540  5368 ?        Ss   08:13   0:00 @dbus-daemo
root         188  0.0  0.1  17980  8888 ?        Ss   08:13   0:00 /usr/lib/sy
root         190  0.0  0.1 1755848 11548 ?       Ssl  08:13   0:00 /usr/libexe
root         209  0.0  0.0   3124  1972 tty1     Ss+  08:13   0:00 /sbin/agett
syslog       210  0.0  0.0 222516  6028 ?        Ssl  08:13   0:00 /usr/sbin/r
root         220  0.0  0.3 107032 23268 ?        Ssl  08:13   0:00 /usr/bin/py
root         307  0.0  0.0   3188  1108 ?        Ss   08:13   0:00 /init
root         308  0.0  0.0   3204  1248 ?        S    08:13   0:00 /init
krishna+     309  0.0  0.0   6080  5420 pts/0    Ss   08:13   0:00 -bash
root         310  0.0  0.0   6704  4676 pts/1    Ss   08:13   0:00 /bin/login
krishna+     353  0.0  0.1  20316 11504 ?        Ss   08:13   0:00 /usr/lib/sy
krishna+     354  0.0  0.0  21168  3596 ?        S    08:13   0:00 (sd-pam)
krishna+     378  0.0  0.0   6080  5368 pts/1    S+   08:13   0:00 -bash
krishna+     715  0.0  0.0   8288  4316 pts/0    R+   08:55   0:00 ps aux
```

```
krishnasree08@krishnasree08:~/task1$ ps --forest
    PID TTY          TIME CMD
    309 pts/0    00:00:00 bash
    716 pts/0    00:00:00  \_ ps
krishnasree08@krishnasree08:~/task1$ top
top - 08:55:33 up 42 min,  1 user,  load average: 0.00, 0.01, 0.00
Tasks:  22 total,   1 running,  21 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0
MiB Mem :   6849.7 total,   6153.8 free,    556.9 used,    286.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   6292.7 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+
      1 root      20   0   21532  12964   9632 S   0.0   0.2   0:00.77
      2 root      20   0    3180   2204   2072 S   0.0   0.0   0:00.01
      6 root      20   0    3196   2128   2008 S   0.0   0.0   0:00.00
     52 root      19  -1   66836  15872  14700 S   0.0   0.2   0:00.43
    100 root      20   0   25284   6640   5152 S   0.0   0.1   0:00.73
    116 systemd+  20   0   21344  13192  11068 S   0.0   0.2   0:00.13
    117 systemd+  20   0   91036   7988   7020 S   0.0   0.1   0:00.15
    180 root      20   0    4244   2688   2432 S   0.0   0.0   0:00.01
    181 message+  20   0    9540   5368   4672 S   0.0   0.1   0:00.15
    188 root      20   0   17980   8888   7848 S   0.0   0.1   0:00.11
    190 root      20   0 1755848  11556   9124 S   0.0   0.2   0:00.20
    209 root      20   0    3124   1972   1836 S   0.0   0.0   0:00.01
    210 syslog    20   0  222516   6028   4640 S   0.0   0.1   0:00.09
    220 root      20   0  107032  23268  13808 S   0.0   0.3   0:00.11
    307 root      20   0    3188   1108    980 S   0.0   0.0   0:00.00
    308 root      20   0    3204   1248   1108 S   0.0   0.0   0:00.06
    309 krishna+  20   0    6080   5420   3704 S   0.0   0.1   0:00.09
    310 root      20   0    6704   4676   3888 S   0.0   0.1   0:00.01
    353 krishna+  20   0   20316  11504   9384 S   0.0   0.2   0:00.11
    354 krishna+  20   0   21168   3596   1868 S   0.0   0.1   0:00.00
    378 krishna+  20   0    6080   5368   3704 S   0.0   0.1   0:00.02
    719 krishna+  20   0    9280   5660   3496 R   0.0   0.1   0:00.01
```

```
krishnasree08@krishnasree08:~/task1$ free -h
               total        used        free      shared  buff/cache   available
Mem:           6.7Gi       553Mi       6.0Gi       3.5Mi       286Mi       6.1Gi
Swap:          2.0Gi          0B       2.0Gi
krishnasree08@krishnasree08:~/task1$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.4G     0  3.4G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.4G  4.0K  3.4G   1% /mnt/wsl
drivers         475G  124G  351G  27% /usr/lib/wsl/drivers
/dev/sdd       1007G  2.0G  954G   1% /
none            3.4G   44K  3.4G   1% /mnt/wslg
none            3.4G     0  3.4G   0% /usr/lib/wsl/lib
rootfs          3.4G  2.8M  3.4G   1% /init
none            3.4G  532K  3.4G   1% /run
none            3.4G     0  3.4G   0% /run/lock
none            3.4G     0  3.4G   0% /run/shm
none            3.4G  100K  3.4G   1% /mnt/wslg/versions.txt
none            3.4G  100K  3.4G   1% /mnt/wslg/doc
C:\             475G  124G  351G  27% /mnt/c
tmpfs           685M   20K  685M   1% /run/user/1000
krishnasree08@krishnasree08:~/task1$ ls /dev | head
autofs
block
bsg
btrfs-control
char
console
core
cpu_dma_latency
cuse
disk
krishnasree08@krishnasree08:~/task1$ cat /proc/meminfo | head
MemTotal:        7014044 kB
MemFree:         6302484 kB
MemAvailable:    6445108 kB
Buffers:           10408 kB
Cached:           260040 kB
SwapCached:            0 kB
Active:            73348 kB
Inactive:         294756 kB
Active(anon):       2680 kB
```

```
krishnasree08@krishnasree08:~/task1$ cat /proc/cpuinfo | head -20
processor       : 0
vendor_id       : AuthenticAMD
cpu family      : 25
model           : 80
model name      : AMD Ryzen 7 7730U with Radeon Graphics
stepping        : 0
microcode       : 0xffffffff
cpu MHz         : 1996.205
cache size      : 512 KB
physical id     : 0
siblings        : 16
core id         : 0
cpu cores       : 8
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 13
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rd
tscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popc
nt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp
 vmmcall fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrip_sav
e tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip vaes vpclmulqdq rdpid fsrm
krishnasree08@krishnasree08:~/task1$
```

Thus, the relationship between hardware resources and operating system services was successfully investigated using Linux terminal commands. The role of Linux in managing CPU, memory, storage, processes and I/O devices was studied and understood.

Summary: In this task, different Linux commands were used to study the hardware resources and operating system services. Commands such as uname, lscpu, and lsblk were used to examine the operating system, CPU, and storage devices. ps and top were used to observe running processes, while free and df helped examine memory and storage usage. The /dev and /proc directories were also studied to understand how Linux provides abstractions for hardware and system resources.